

Project Design Phase

Solution Architecture

Date	27 July 2026	Team ID	Not Specified
Project Name	Automated Network Request Management	Maximum Marks	4 Marks

The proposed solution architecture leverages the **ServiceNow platform** to automate the complete lifecycle of network service requests. Employees submit requests through the **Service Catalog**, where request details are captured using catalog variables. **Flow Designer** automates record creation, approvals, task assignment, notifications, and request tracking. All request information is stored in a centralized **Network Request** table, enabling administrators to monitor requests through reports and dashboards while ensuring a standardized and efficient request management process.

Architecture and Data Flow

Employee / User Submits a Network Access Request by selecting the request type, device type, access level, business justification, and requested user.	Service Catalog Provides a centralized self-service interface to collect standardized network request information.	Flow Designer Validates catalog variables, creates the Network Request record, routes requests for approval, assigns tasks, and automates the workflow.
Approval Engine Routes the request to the Manager or Network Administrator for approval or rejection.	Ticket Assignment Automatically assigns approved requests to the appropriate IT Support Group based on the request type.	Notifications Sends automated notifications for request submission, approval, rejection, assignment, and completion..
Network Request Record Stores request details, approval status, assignment information, request history, and current status.	Reports & Dashboards Provide administrators with real-time visibility into request volume, approvals, pending requests, completed requests, and operational performance.	Administration Allows administrators to monitor, manage, update, and maintain the complete network request lifecycle.

Component Description

Component	Function
Employee / User	Initiates the network service request by submitting required details through the Service Catalog.
Service Catalog	Provides the request interface and captures standardized network request information using catalog variables.
Catalog Variables	Collects request type, device type, access level, priority, business justification, and requested user information.
Flow Designer	Automates request validation, Network Request record creation, approvals, task assignment, and workflow execution.
Network Request Table	Stores all network request records and request-related information in a centralized database.

Approval Engine	Routes requests to the appropriate manager or network administrator for approval.
Task Assignment	Automatically assigns approved requests to the correct IT support group for implementation.
Notifications	Sends automated email or system notifications throughout the request lifecycle.
Reports & Dashboards	Displays request statistics, approval status, pending requests, completed requests, and overall system performance.
System Administrator	Configures, monitors, maintains, and manages the complete application and workflow.

End-to-End Process Flow

1. Employee opens the **Network Access Request** catalog item.
2. Employee enters request details including request type, device type, access level, priority, business justification, and requested user.
3. The **Service Catalog** captures the submitted request.
4. **Flow Designer** retrieves the catalog variables.
5. The flow validates the submitted information.
6. A **Network Request** record is automatically created in the custom table.
7. The request is routed to the Manager or Network Administrator for approval.
8. After approval, the request is automatically assigned to the appropriate IT Support Group.
9. Notifications are sent to the requester and assigned support team.
10. IT Support processes the request and updates the request status.
11. Reports and dashboards are updated to provide real-time monitoring and analytics.
12. The request is completed and the employee receives a completion notification.

Technology Stack

Technology	Usage
Platform	ServiceNow (Australia Release)
Development Environment	ServiceNow Studio
Application Development	Scoped Application
User Interface	Service Catalog
Automation	Flow Designer
Workflow	Approval Engine
Database	Custom Network Request Table
Forms	Form Designer
Data Collection	Catalog Variables
Notifications	ServiceNow Notifications

Technology**Usage****Reporting**

ServiceNow Reports

Analytics

Dashboards

Security

Roles & ACLs

Assignment

Assignment Groups

Programming / Scripting

Business Rules, Client Scripts, UI Policies, Script Includes (Future Implementation)